

GVT Quick Tour

Note: this is an alpha version of GVT. It is based on the well-known Python version of GVT. Functionality is not complete. It is included for testing purposes.

Preliminaries

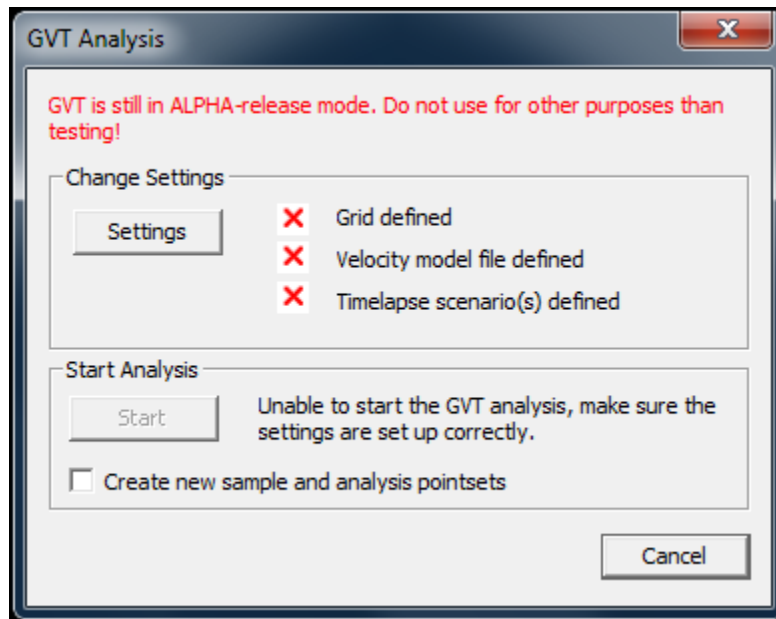
You need a model with results. If Non-linear results are available, GVT will use those. Otherwise it will default to Linear results.

Background information about GVT is available in Geomec's Help menu, under Background theory->GVT Analysis.

If you are familiar with the Python version of GVT, please visit the section Differences with the Python GVT.

Setting up GVT analysis

GVT Analysis is available under the Analysis menu, which leads to this dialog:



Press "Change Settings" to set up GVT Analysis. If there are no red crosses but only green check marks, you may skip this step (although it can never hurt to check your settings).

GVT Settings



GVT Input

Survey Boxes

grid0

New Box

Edit Box

Delete Box

Initial Velocity Model

☐ File

Browse

☒ Take the initial velocities from the Geomec material models (Vp)

OK

Cancel

GVT Output

Output prefix:

Run1_

Output directory:

C:\Apps

Browse

☒ Delta t

☒ Delta v

☒ Import back to Geomec

Velocity Model Parameter Definition

Formations	Velocity Model	Parameter	Value
Baserock	R-Factor	R_C	1
Inclusion	R-Factor	R_EX	5
LowerReservoir FB	R-Factor		
TopReservoir	R-Factor		
shallowcaprock	R-Factor		
TopResFB	R-Factor		
Deepcaprock	R-Factor		
shallowoverburden	R-Factor		
Deepoverburden	R-Factor		

Time-Lapse Scenarios

Scenario	Depl. Stage 1	Depl. Stage 2	
Scenario 0	Initial	Depletion stage 1	

Add

Remove

The Settings dialog follows a natural order from top to bottom, so it is convenient to make a New Box first. A new dialog will open:

The image shows a software dialog box titled "GVT Survey Box Definition". It contains several input fields and buttons for configuring a survey box. The fields are organized into sections: "Name", "Import 123DI Keyfile", "Manual override", "Corner Point", "Azimuth", "Maximum Depth", "Bin / Track Spacing", "First Bin / Track", and "Number of Bin / Track". The "Name" field contains "grid0". The "Keyfile" field is empty, with a "Browse" button next to it. The "Corner Point" section has "Northing" set to 2354300 and "Easting" set to 507031, both in meters. The "Azimuth" field is set to 35 degrees. The "Maximum Depth" field is set to 1200.00024414063 meters. The "Bin / Track Spacing" section has "Bin spacing" and "Track spacing" both set to 200 meters. The "First Bin / Track" section has "First bin" and "First track" both set to 0. The "Number of Bin / Track" section has "Number of bin" set to 35 and "Number of track" set to 25. On the right side of the dialog, there are "OK" and "Cancel" buttons.

GVT Survey Box Definition

Name: grid0

Import 123DI Keyfile: Keyfile: [] Browse

Manual override

Corner Point

Northing: 2354300 m

Easting: 507031 m

Azimuth

Azimuth: 35 deg

Maximum Depth

Depth: 1200.00024414063 m

Bin / Track Spacing

Bin spacing: 200 m

Track spacing: 200 m

First Bin / Track

First bin: 0

First track: 0

Number of Bin / Track

Number of bin: 35

Number of track: 25

OK

Cancel

The “Name” is important, as it is the prefix of all the data sets that GVT generates, or re-uses.

You can either import a 123DI keyfile, and override certain options, or skip this and just use the manual options. Almost all options are available in the “Manual override”. If you need many analyses with the same Survey Box, you might want to create a keyfile. See Appendix A for an example.

The “Corner Point” is the start of your grid coordinate system. This coordinate system can be rotated with “Azimuth”, the step size on the axes can be set with “Bin / Track Spacing” and how far the coordinate system extends with “Number of Bin / Track”.

The actual first point of your grid is defined by “First Bin / Track”, and from there it will fill up the coordinate system.

The order of Bin and Track follow the order of the “Corner Point” definition: Bins are in the Northing direction, Tracks in the Easting direction.

“Maximum Depth” is not used in the alpha release.

With everything filled in, press OK to return to the Settings dialog.

In the Settings dialog, the “Initial Velocity” should be set to “Take the velocities from the Geomec material models (Vp)”.

Velocity files are not yet supported in the alpha release. In the future, import of velocity files as Vp data sets in Geomec will be possible. If a velocity file is needed, this conversion and import needs to be done manually.

“GVT Output” is not supported. Like the velocity files, the export will be added in the future. Right now, export and conversion of the output data sets can be done manually.

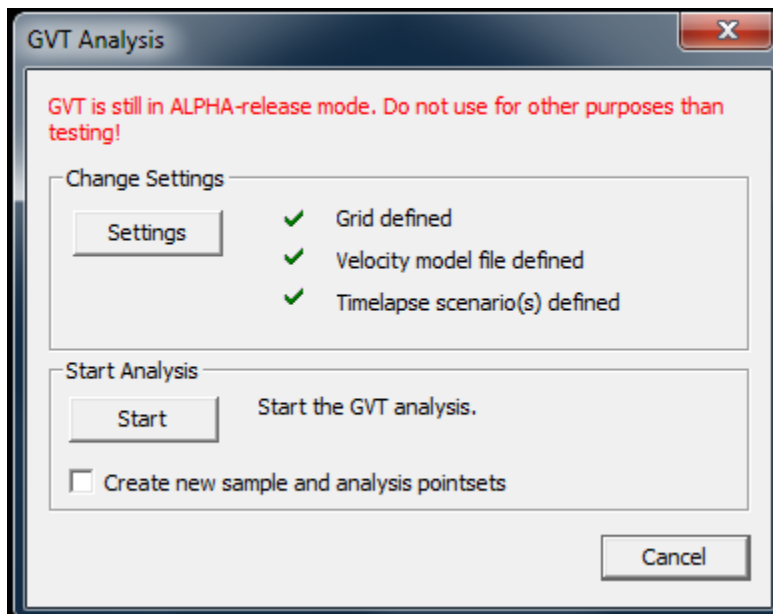
As for choosing between ‘Delta t’ and ‘Delta v’, both are calculated at the same time, without loss of performance.

In the “Velocity Model Parameter Definition” you can choose which model to use for each formation, and optionally change the values (per formation) in the right tab. See the Help background information for details.

At least one “Time-lapse Scenario” needs to be added, from Initial stage to a later stage.

Press OK to return to the Analysis dialog.

Running GVT Analysis



If all is well, the Analysis dialog now shows green check marks. If not, please go to the Settings dialog again, and check if you missed something.

Also note that you can Cancel now, and the Settings will be there if your return to this dialog. You may want to check if any of the generated pointsets (prefixed with the Survey Box/Grid "Name") already exist. If they do, GVT will re-use them. If that is not your intent, you can remove them from the Data Storage, or check "Create new sample and analysis pointsets" here.

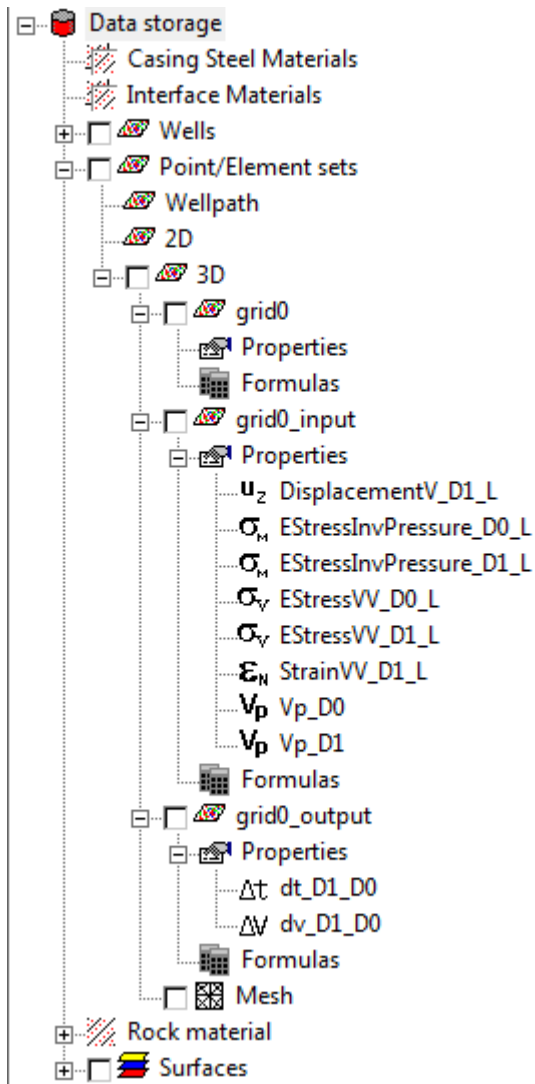
If you are all set, press "Start" to begin the analysis.

GVT will first create an input grid, then two data set with "pillar maps" (the same grid in 3D, with all the points on where the pillar intersects with the element sides). The first set is the input set, taken from the model. This is present for testing purposes. The second set is the output set, and the result of the GVT Analysis.

In practice, building the input sets might take far longer than the actual analysis, and the latter may be fast enough to escape notice. Once the progress bars disappear, check the Data storage for the existence of the three different data sets.

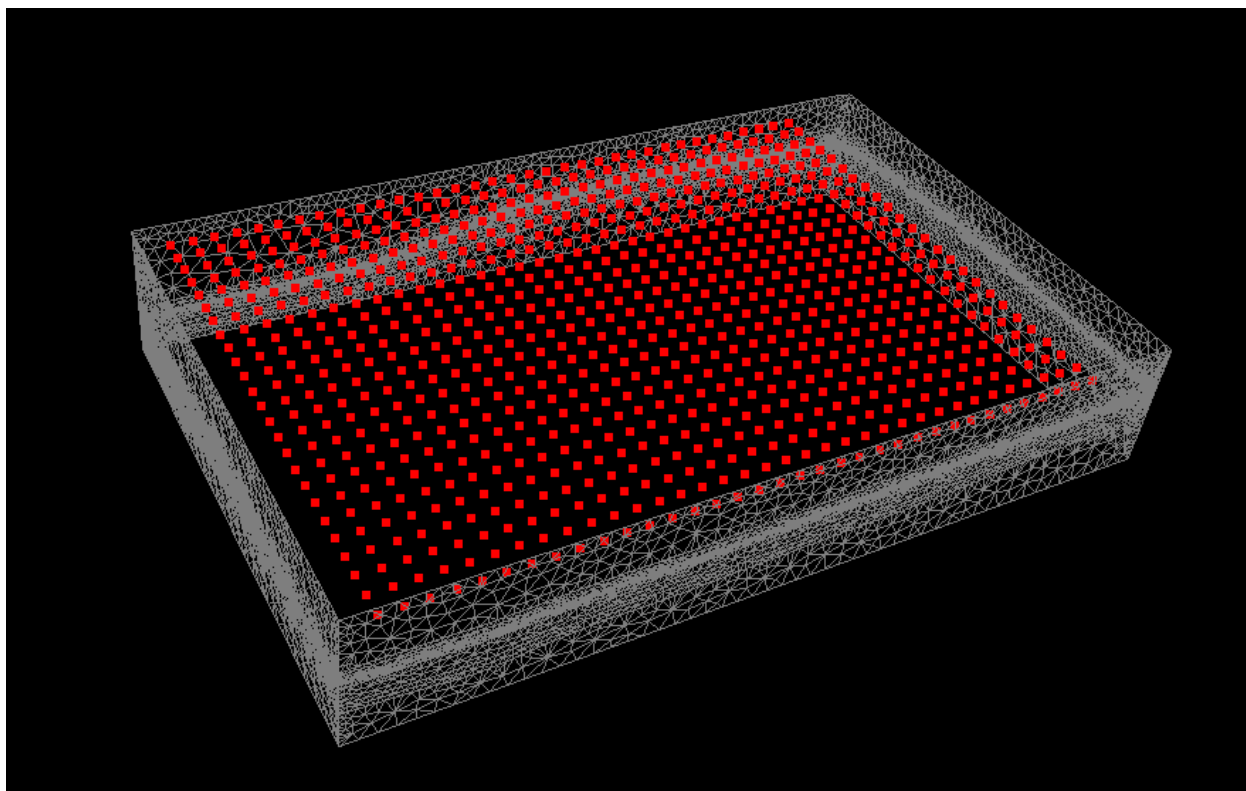
GVT Output

The GVT Analysis should add three data sets to the Data Storage:

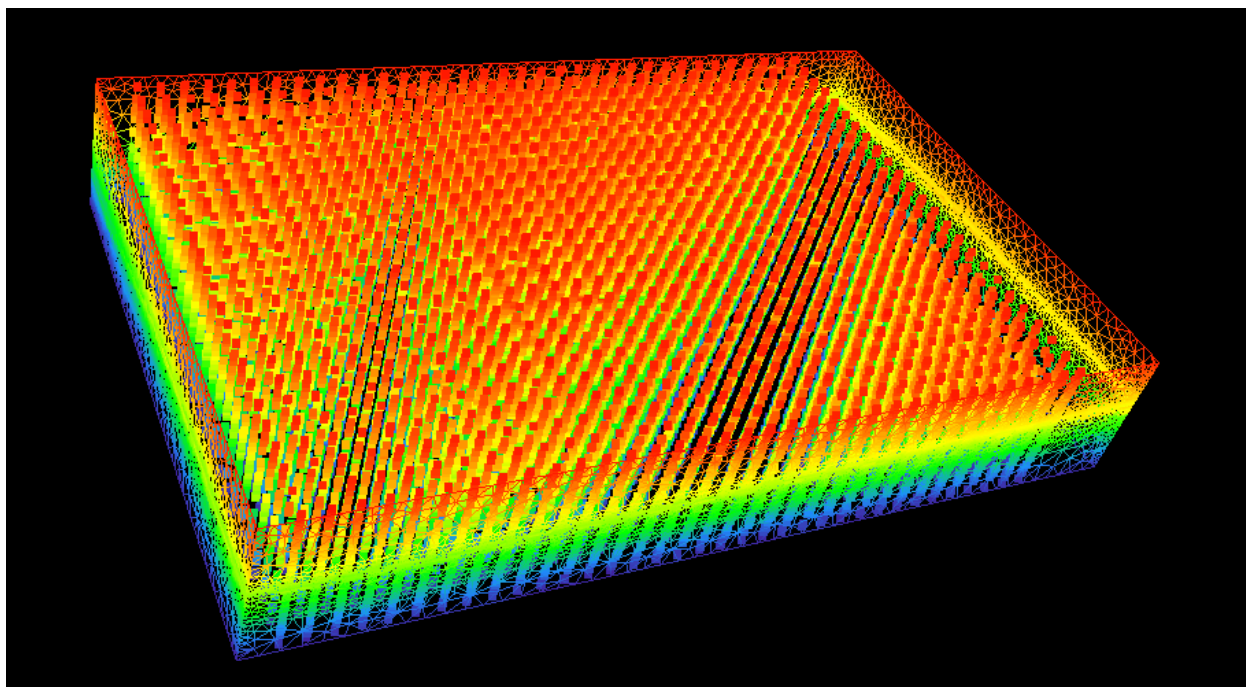


"grid0" is our 2D grid, "grid0_input" our input set, and "grid0_output" the result of the calculation.

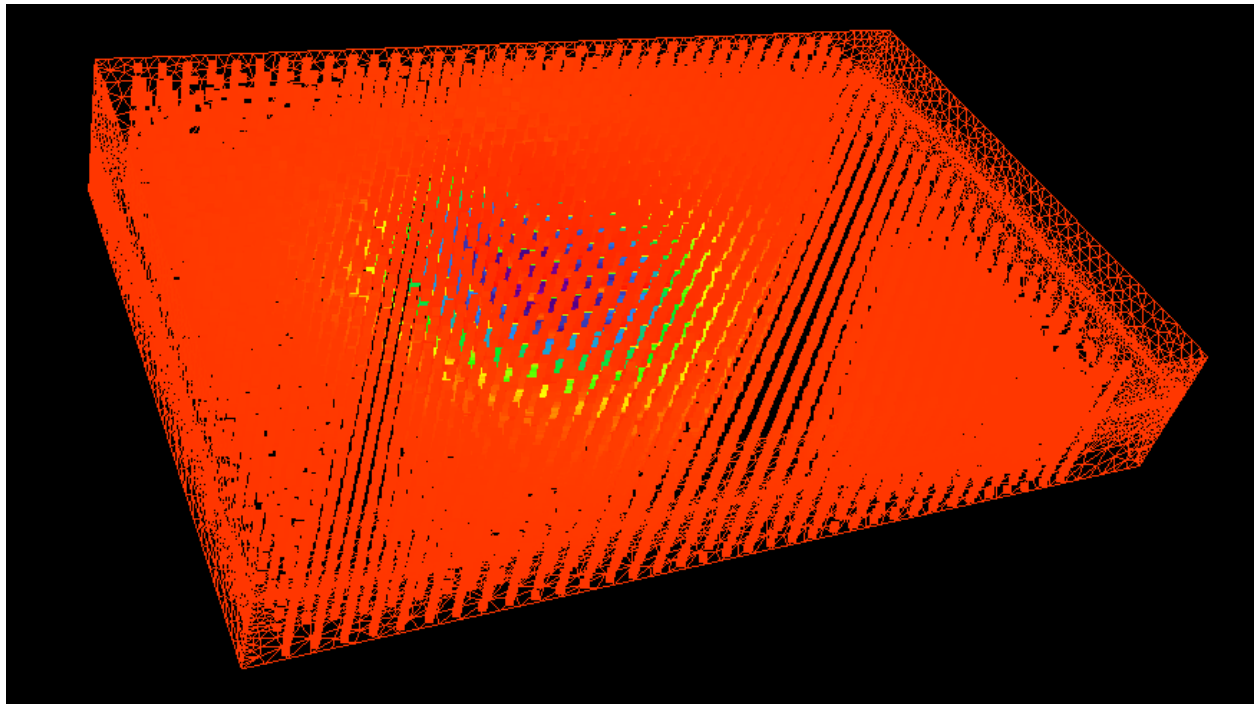
Below are pictures of each set, viewed together with the model boundary.



The 2D “grid0”.



The “grid0_input” pillar map.



The “grid0_output” pillar map.

Appendix A. Keyfile example

```
MESSAGE_FILE

DELTA_BIN_NUM I 1

DELTA_TRACK_NUM I 1

FIRST_BIN I 0

FIRST_TRACK I 0

N_BIN I 35

N_TRACK I 25

REAL_DELTA_X R 200

REAL_DELTA_Y R 200

REAL_GEO_X R 2354300

REAL_GEO_Y R 507031

SKEW_ANGLE R 35
```

More fields are possible, but they are not handled by Geomec's GVT.

DELTA_BIN_NUM and DELTA_TRACK_NUM cannot be set via the Survey Box dialog, but you can set them in the keyfile, and Geomec will use them. They can be negative, or positive, and determine in which direction your coordinate system extends.